

# ESP32-MINI-1

# ESP32-MINI-1U

## Datasheet

2.4 GHz Wi-Fi + Bluetooth® + Bluetooth LE module

Built around ESP32 series of SoC, Xtensa® dual-core 32-bit LX6 microprocessor

4 MB flash

28 GPIOs, rich set of peripherals

On-board PCB antenna or external antenna connector



ESP32-MINI-1



ESP32-MINI-1U



Version 1.3  
Espressif Systems  
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# 1 Module Overview

**Note:**

Check the link or the QR code to make sure that you use the latest version of this document:

[https://espressif.com/sites/default/files/documentation/esp32-mini-1\\_datasheet\\_en.pdf](https://espressif.com/sites/default/files/documentation/esp32-mini-1_datasheet_en.pdf)



## 1.1 Features

### CPU and On-Chip Memory

- ESP32-U4WDH embedded, Xtensa dual-core 32-bit LX6 microprocessor, up to 240 MHz
- 448 KB ROM
- 520 KB SRAM
- 16 KB SRAM in RTC
- 4 MB SPI flash

### Wi-Fi

- 802.11b/g/n
- Bit rate: 802.11n up to 150 Mbps
- A-MPDU and A-MSDU aggregation
- 0.4  $\mu$ s guard interval support
- Center frequency range of operating channel: 2412 ~ 2484 MHz

### Bluetooth

- Bluetooth v4.2 BR/EDR and Bluetooth LE specification
- Class-1, class-2, and class-3 transmitter
- AFH
- CVSD and SBC

### Peripherals

- SD card, UART, SPI, SDIO, I2C, LED PWM, motor PWM, I2S, infrared remote controller, pulse counter, GPIO, touch sensor, ADC, DAC, Ethernet, TWAI® (compatible with ISO 11898-1, i.e. CAN Specification 2.0)

### Integrated Components on Module

- 40 MHz crystal oscillator

### Antenna Options

- ESP32-MINI-1: On-board PCB antenna
- ESP32-MINI-1U: external antenna via a connector

### Operating Conditions

- Operating voltage/Power supply: 3.0 ~ 3.6 V
- Operating ambient temperature:
  - 85 °C version module: -40 ~ 85 °C
  - 105 °C version module: -40 ~ 105 °C

### Certification

- RF certification: See [Certificates](#)
- Green certification: REACH/RoHS

### Test

- Reliability: HTOL/HTSL/uHAST/TCT/ESD

## 1.2 Description

ESP32-MINI-1 and ESP32-MINI-1U are highly-integrated, small-sized Wi-Fi + Bluetooth + Bluetooth LE MCU modules that have a rich set of peripherals. The modules are ideal choices for a wide variety of IoT applications, ranging from home automation, smart building, consumer electronics to industrial control, especially suitable for applications within a compact space, such as bulbs, switches and sockets.

ESP32-MINI-1 comes with a on-board PCB antenna, and ESP32-MINI-1U with a connector for external antenna. Both ESP32-MINI-1 and ESP32-MINI-1U have two variants:

- 85 °C version: integrating the ESP32-U4WDH chip
- 105 °C version: integrating the ESP32-U4WDH chip

In this datasheet unless otherwise stated, ESP32-MINI-1 refers to both ESP32-MINI-1-N4 and ESP32-MINI-1-H4, whereas ESP32-MINI-1U refers to both ESP32-MINI-1U-N4 and ESP32-MINI-1U-H4.

The ordering information for ESP32-MINI-1 and ESP32-MINI-1U is listed as follows:

**Table 1: Ordering Information**

| Module                         | Ordering code          | Chip embedded | Module dimensions (mm) |
|--------------------------------|------------------------|---------------|------------------------|
| ESP32-MINI-1 (85 °C version)   | <b>ESP32-MINI-1-N4</b> | ESP32-U4WDH   | 13.2 × 19.0 × 2.4      |
| ESP32-MINI-1 (105 °C version)  | ESP32-MINI-1-H4        | ESP32-U4WDH   |                        |
| ESP32-MINI-1U (85 °C version)  | ESP32-MINI-1U-N4       | ESP32-U4WDH   | 13.2 × 13.5 × 2.4      |
| ESP32-MINI-1U (105 °C version) | ESP32-MINI-1U-H4       | ESP32-U4WDH   |                        |

At the core of this module is ESP32-U4WDH\*, an Xtensa 32-bit LX6 CPU that operates at up to 240 MHz. The user can power off the CPU and make use of the low-power co-processor to constantly monitor the peripherals for changes or exceeding of thresholds.

This ESP32 chip integrates a rich set of peripherals, ranging from capacitive touch sensor, SD card interface, Ethernet, high-speed SPI, UART, I2S, I2C, etc.

For more information on ESP32 chips, please refer to [ESP32 Series Datasheet](#).

## 1.3 Applications

- Generic Low-power IoT Sensor Hub
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming
- Over-the-top (OTT) Devices
- Speech Recognition
- Image Recognition
- Mesh Network
- Home Automation
- Smart Building
- Industrial Automation
- Smart Agriculture
- Audio Applications
- Health Care Applications
- Wi-Fi-enabled Toys
- Wearable Electronics
- Retail & Catering Applications

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## 2 Block Diagram

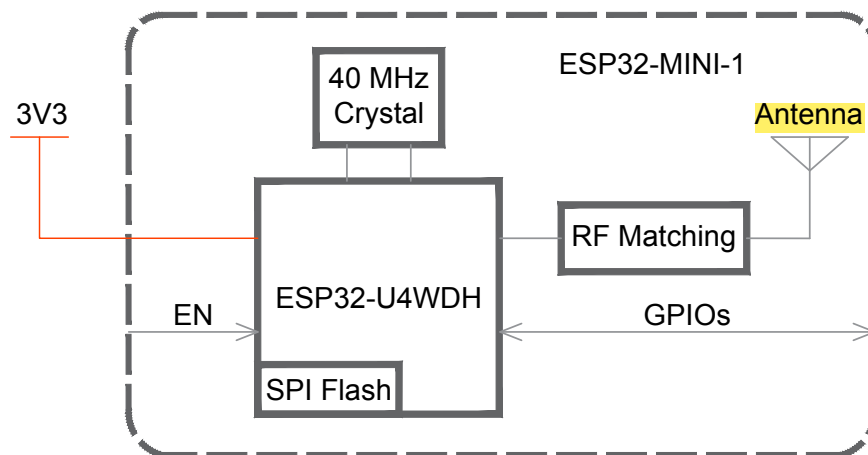


Figure 1: ESP32-MINI-1 Block Diagram

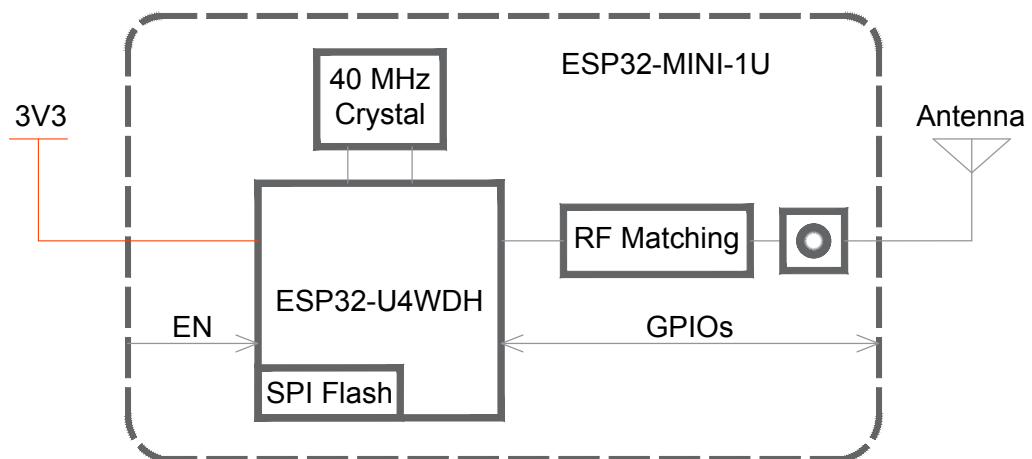


Figure 2: ESP32-MINI-1U Block Diagram

## 3 Pin Definitions

### 3.1 Pin Layout

The pin diagrams below show the approximate location of pins on the module. For the actual diagram drawn to scale, please refer to Figure 7.1 *Physical Dimensions*.

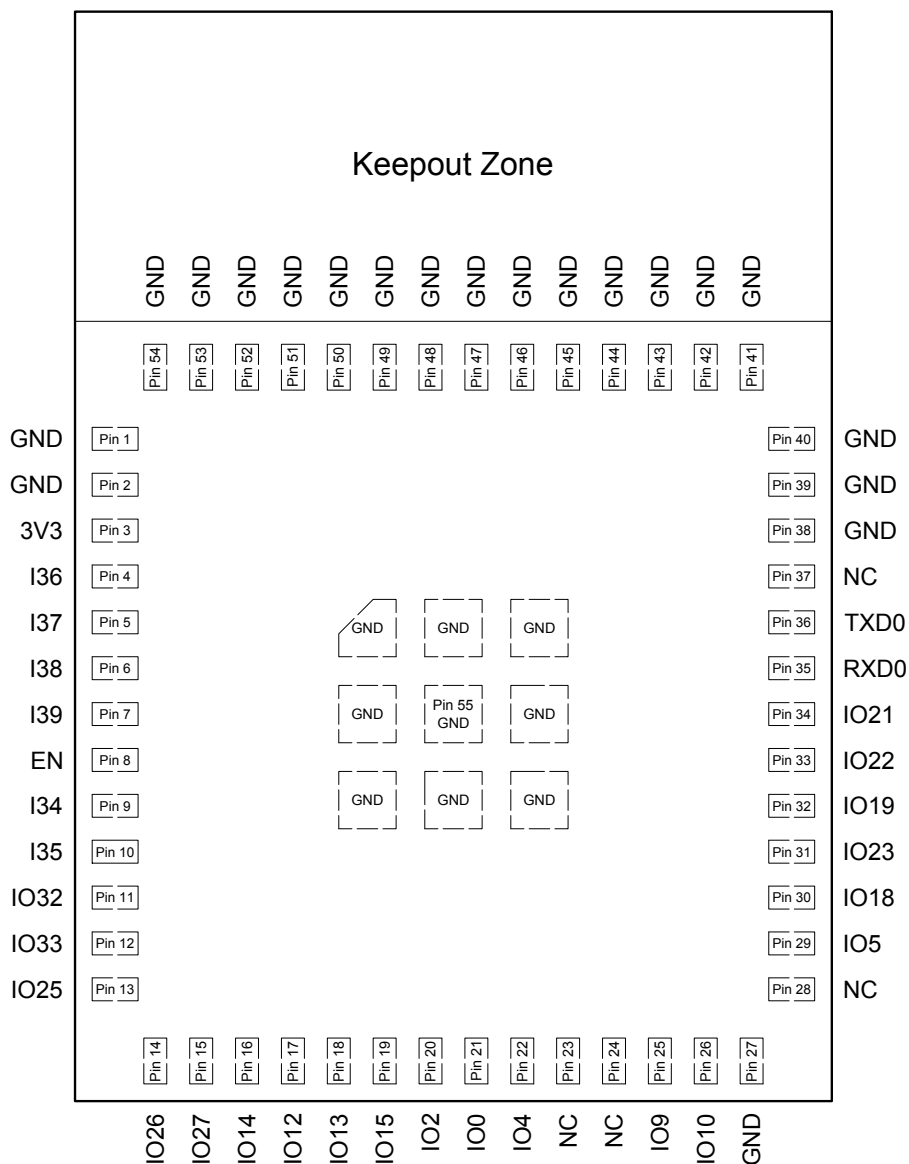
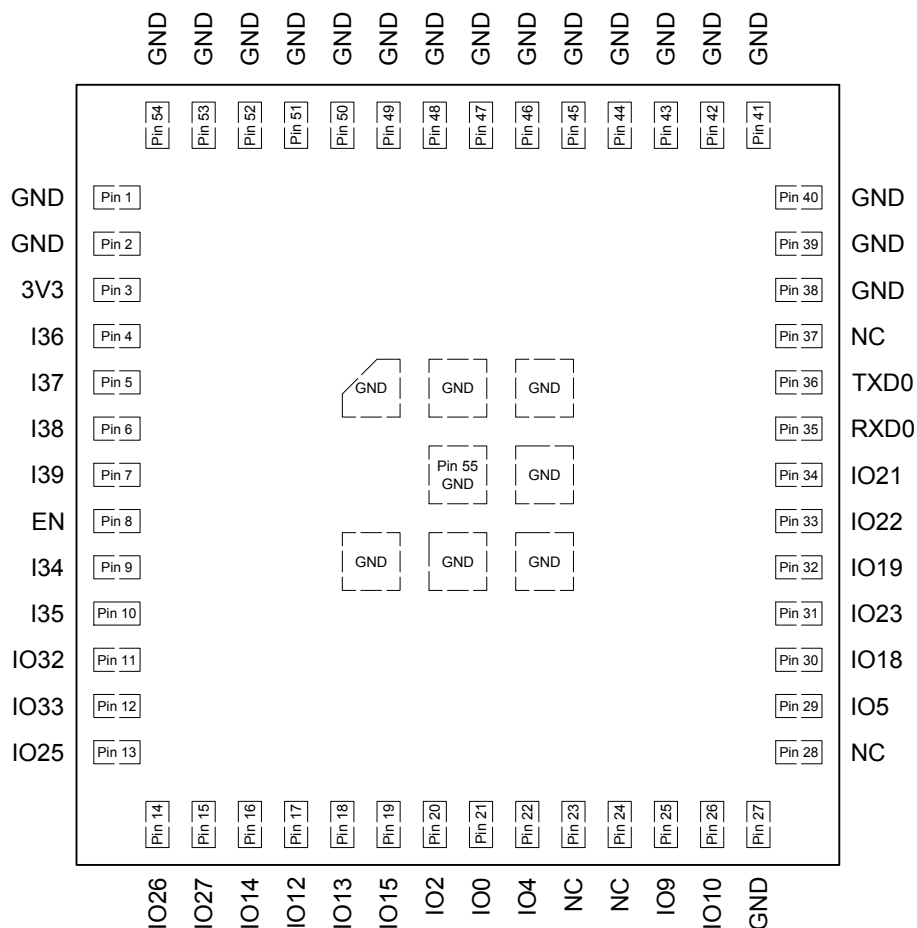


Figure 3: ESP32-MINI-1 Pin Layout (Top View)



### Figure 4: ESP32-MINI-1U Pin Layout (Top View)

### 3.2 Pin Description

ESP32-MINI-1 and ESP32-MINI-1U each has 55 pins. See pin definitions in Table 2. For peripheral pin configurations, please refer to [ESP32 Series Datasheet](#).

### Table 2: Pin Definitions

| Name | No.               | Type* | Function                                                                                  |
|------|-------------------|-------|-------------------------------------------------------------------------------------------|
| GND  | 1, 2, 27, 38 ~ 55 | P     | Ground                                                                                    |
| 3V3  | 3                 | P     | Power supply                                                                              |
| I36  | 4                 | I     | GPIO36, ADC1_CH0, RTC_GPIO0                                                               |
| I37  | 5                 | I     | GPIO37, ADC1_CH1, RTC_GPIO1                                                               |
| I38  | 6                 | I     | GPIO38, ADC1_CH2, RTC_GPIO2                                                               |
| I39  | 7                 | I     | GPIO39, ADC1_CH3, RTC_GPIO3                                                               |
| EN   | 8                 | I     | High: enables the chip<br>Low: the chip shuts down<br>Note: do not leave the pin floating |
| I34  | 9                 | I     | GPIO34, ADC1_CH6, RTC_GPIO4                                                               |
| I35  | 10                | I     | GPIO35, ADC1_CH7, RTC_GPIO5                                                               |

Cont'd on next page

Table 2 – cont'd from previous page

| Name | No. | Type | Function                                                                               |
|------|-----|------|----------------------------------------------------------------------------------------|
| IO32 | 11  | I/O  | GPIO32, XTAL_32K_P (32.768 kHz crystal oscillator input), ADC1_CH4, TOUCH9, RTC_GPIO9  |
| IO33 | 12  | I/O  | GPIO33, XTAL_32K_N (32.768 kHz crystal oscillator output), ADC1_CH5, TOUCH8, RTC_GPIO8 |
| IO25 | 13  | I/O  | GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0                                          |
| IO26 | 14  | I/O  | GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1                                          |
| IO27 | 15  | I/O  | GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV                                       |
| IO14 | 16  | I/O  | GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMS, HSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2        |
| IO12 | 17  | I/O  | GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3      |
| IO13 | 18  | I/O  | GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER     |
| IO15 | 19  | I/O  | GPIO15, ADC2_CH3, TOUCH3, RTC_GPIO13, MTDO, HSPICS0, HS2_CMD, SD_CMD, EMAC_RXD3        |
| IO2  | 20  | I/O  | GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPWP, HS2_DATA0, SD_DATA0                        |
| IO0  | 21  | I/O  | GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK                             |
| IO4  | 22  | I/O  | GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER            |
| NC   | 23  | -    | No connect                                                                             |
| NC   | 24  | -    | No connect                                                                             |
| IO9  | 25  | I/O  | GPIO9, HS1_DATA2, U1RXD, SD_DATA2                                                      |
| IO10 | 26  | I/O  | GPIO10, HS1_DATA3, U1TXD, SD_DATA3                                                     |
| NC   | 28  | -    | No connect                                                                             |
| IO5  | 29  | I/O  | GPIO5, HS1_DATA6, VSPICS0, EMAC_RX_CLK                                                 |
| IO18 | 30  | I/O  | GPIO18, HS1_DATA7, VSPICLK                                                             |
| IO23 | 31  | I/O  | GPIO23, HS1_STROBE, VSPID                                                              |
| IO19 | 32  | I/O  | GPIO19, VSPIQ, U0CTS, EMAC_TXD0                                                        |
| IO22 | 33  | I/O  | GPIO22, VSPIWP, U0RTS, EMAC_TXD1                                                       |
| IO21 | 34  | I/O  | GPIO21, VSPIHD, EMAC_TX_EN                                                             |
| RXD0 | 35  | I/O  | GPIO3, U0RXD, CLK_OUT2                                                                 |
| TXD0 | 36  | I/O  | GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2                                                      |
| NC   | 37  | -    | No connect                                                                             |

\* Pins GPIO6, GPIO7, GPIO8, GPIO11, GPIO16, and GPIO17 on the ESP32-U4WDH chip are connected to the SPI flash integrated in the chip and cannot be used for other purposes except for connecting external PSRAM.

\* P: power supply; I: input; O: output.

### 3.3 Strapping Pins

**Note:**

The content below is excerpted from Section Strapping Pins in [ESP32 Series Datasheet](#). For the strapping pin mapping between the chip and modules, please refer to Chapter 5 [Module Schematics](#).

ESP32 has five strapping pins:

- MTDI
- GPIO0
- GPIO2
- MTDO
- GPIO5

Software can read the values of these five bits from register "GPIO\_STRAPPING".

During the chip's system reset release (power-on-reset, RTC watchdog reset and brownout reset), the latches of the strapping pins sample the voltage level as strapping bits of "0" or "1", and hold these bits until the chip is powered down or shut down. The strapping bits configure the device's boot mode, the operating voltage of VDD\_SDIO and other initial system settings.

Each strapping pin is connected to its internal pull-up/pull-down during the chip reset. Consequently, if a strapping pin is unconnected or the connected external circuit is high-impedance, the internal weak pull-up/pull-down will determine the default input level of the strapping pins.

To change the strapping bit values, users can apply the external pull-down/pull-up resistances, or use the host MCU's GPIOs to control the voltage level of these pins when powering on ESP32.

After reset release, the strapping pins work as normal-function pins.

Refer to Table 3 for a detailed boot-mode configuration by strapping pins.

**Table 3: Strapping Pins**

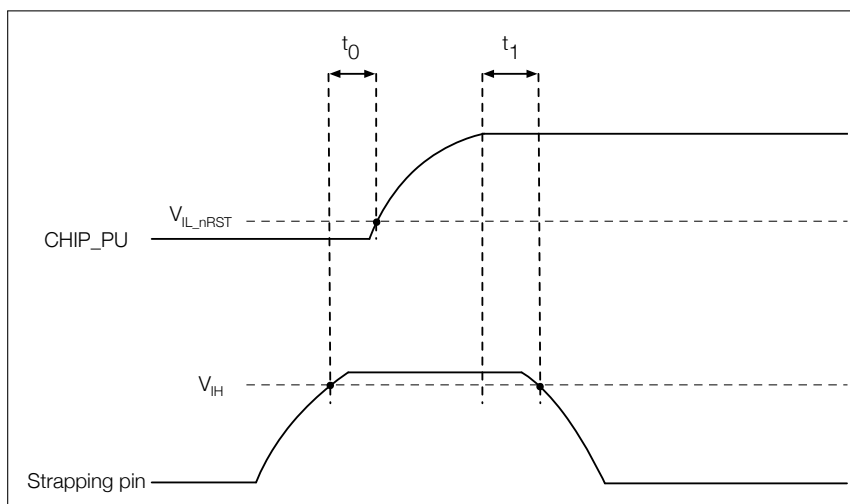
| Voltage of Internal LDO (VDD_SDIO)                                |           |                          |                          |                          |                          |
|-------------------------------------------------------------------|-----------|--------------------------|--------------------------|--------------------------|--------------------------|
| Pin                                                               | Default   | 3.3 V                    |                          | 1.8 V                    |                          |
| MTDI                                                              | Pull-down | 0                        |                          | 1                        |                          |
| Bootling Mode                                                     |           |                          |                          |                          |                          |
| Pin                                                               | Default   | SPI Boot                 |                          | Download Boot            |                          |
| GPIO0                                                             | Pull-up   | 1                        |                          | 0                        |                          |
| GPIO2                                                             | Pull-down | Don't-care               |                          | 0                        |                          |
| Enabling/Disabling Debugging Log Print over U0TXD During Bootling |           |                          |                          |                          |                          |
| Pin                                                               | Default   | U0TXD Active             |                          | U0TXD Silent             |                          |
| MTDO                                                              | Pull-up   | 1                        |                          | 0                        |                          |
| Timing of SDIO Slave                                              |           |                          |                          |                          |                          |
| Pin                                                               | Default   | FE Sampling<br>FE Output | FE Sampling<br>RE Output | RE Sampling<br>FE Output | RE Sampling<br>RE Output |
| MTDO                                                              | Pull-up   | 0                        | 0                        | 1                        | 1                        |
| GPIO5                                                             | Pull-up   | 0                        | 1                        | 0                        | 1                        |

\* FE: falling-edge, RE: rising-edge

\* Firmware can configure register bits to change the settings of "Voltage of Internal LDO (VDD\_SDIO)" and "Timing of SDIO Slave", after bootling.

\* The module integrates a 3.3 V SPI flash, so the pin MTDI cannot be set to 1 when the module is powered up.

The illustration below shows the setup and hold times for the strapping pins before and after the CHIP\_PU signal goes high. Details about the parameters are listed in Table 4.

**Figure 5: Setup and Hold Times for the Strapping Pins****Table 4: Parameter Descriptions of Setup and Hold Times for the Strapping Pins**

| Parameters | Description                                     | Min. | Unit |
|------------|-------------------------------------------------|------|------|
| $t_0$      | Setup time before CHIP_PU goes from low to high | 0    | ms   |
| $t_1$      | Hold time after CHIP_PU goes high               | 1    | ms   |

## 4 Electrical Characteristics

### 4.1 Absolute Maximum Ratings

Stresses above those listed in *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

**Table 5: Absolute Maximum Ratings**

| Symbol             | Parameter            | Min  | Max | Unit |
|--------------------|----------------------|------|-----|------|
| VDD33              | Power supply voltage | −0.3 | 3.6 | V    |
| T <sub>STORE</sub> | Storage temperature  | −40  | 105 | °C   |

\* Please see Appendix IO MUX of [ESP32 Series Datasheet](#) for IO's power domain.

### 4.2 Recommended Operating Conditions

**Table 6: Recommended Operating Conditions**

| Symbol           | Parameter                                  | Min | Typ | Max | Unit |
|------------------|--------------------------------------------|-----|-----|-----|------|
| VDD33            | Power supply voltage                       | 3.0 | 3.3 | 3.6 | V    |
| I <sub>VDD</sub> | Current delivered by external power supply | 0.5 | —   | —   | A    |
| T                | Operating ambient temperature              | −40 | —   | 85  | °C   |
|                  |                                            |     |     | 105 |      |

### 4.3 DC Characteristics (3.3 V, 25 °C)

**Table 7: DC Characteristics (3.3 V, 25 °C)**

| Symbol          | Parameter                 | Min                     | Typ | Max                     | Unit |
|-----------------|---------------------------|-------------------------|-----|-------------------------|------|
| C <sub>IN</sub> | Pin capacitance           | —                       | 2   | —                       | pF   |
| V <sub>IH</sub> | High-level input voltage  | 0.75 × VDD <sup>1</sup> | —   | VDD <sup>1</sup> + 0.3  | V    |
| V <sub>IL</sub> | Low-level input voltage   | −0.3                    | —   | 0.25 × VDD <sup>1</sup> | V    |
| I <sub>IH</sub> | High-level input current  | —                       | —   | 50                      | nA   |
| I <sub>IL</sub> | Low-level input current   | —                       | —   | 50                      | nA   |
| V <sub>OH</sub> | High-level output voltage | 0.8 × VDD <sup>1</sup>  | —   | —                       | V    |
| V <sub>OL</sub> | Low-level output voltage  | —                       | —   | 0.1 × VDD <sup>1</sup>  | V    |

Cont'd on next page

Table 7 – cont'd from previous page

| Symbol         | Parameter                                                                                                                             |                                            | Min | Typ | Max | Unit |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----|-----|-----|------|
| $I_{OH}$       | High-level source current<br>(VDD <sup>1</sup> = 3.3 V,<br>V <sub>OH</sub> >= 2.64 V,<br>output drive strength set<br>to the maximum) | VDD3P3_CPU<br>power domain <sup>1, 2</sup> | —   | 40  | —   | mA   |
|                |                                                                                                                                       | VDD3P3_RTC<br>power domain <sup>1, 2</sup> | —   | 40  | —   | mA   |
|                |                                                                                                                                       | VDD_SDIO power<br>domain <sup>1, 3</sup>   | —   | 20  | —   | mA   |
| $I_{OL}$       | Low-level sink current<br>(VDD <sup>1</sup> = 3.3 V, V <sub>OL</sub> = 0.495 V,<br>output drive strength set to the maximum)          |                                            | —   | 28  | —   | mA   |
| $R_{PU}$       | Resistance of internal pull-up resistor                                                                                               |                                            | —   | 45  | —   | kΩ   |
| $R_{PD}$       | Resistance of internal pull-down resistor                                                                                             |                                            | —   | 45  | —   | kΩ   |
| $V_{IL\_nRST}$ | Low-level input voltage of CHIP_PU<br>to shut down the chip                                                                           |                                            | —   | —   | 0.6 | V    |

<sup>1</sup> Please see Appendix IO MUX of [ESP32 Series Datasheet](#) for IO's power domain. VDD is the I/O voltage for a particular power domain of pins.

<sup>2</sup> For VDD3P3\_CPU and VDD3P3\_RTC power domain, per-pin current sourced in the same domain is gradually reduced from around 40 mA to around 29 mA,  $V_{OH} \geq 2.64\text{ V}$ , as the number of current-source pins increases.

<sup>3</sup> Pins occupied by flash and/or PSRAM in the VDD\_SDIO power domain were excluded from the test.

## 4.4 Current Consumption Characteristics

Owing to the use of advanced power-management technologies, the module can switch between different power modes. For details on different power modes, please refer to Section *RTC and Low-Power Management* in [ESP32 Series Datasheet](#).

Table 8: Current Consumption Depending on RF Modes

| Work Mode           | Description     |                                    | Peak (mA) |
|---------------------|-----------------|------------------------------------|-----------|
| Active (RF working) | TX              | 802.11b, 20 MHz, 1 Mbps, @19.5 dBm | 379       |
|                     |                 | 802.11g, 20 MHz, 54 Mbps, @15 dBm  | 276       |
|                     |                 | 802.11n, 20 MHz, MCS7, @13 dBm     | 258       |
|                     |                 | 802.11n, 40 MHz, MCS7, @13 dBm     | 260       |
|                     | RX <sup>2</sup> | 802.11b/g/n, 20 MHz                | 112       |
|                     |                 | 802.11n, 40 MHz                    | 118       |

<sup>1</sup> The current consumption measurements are taken with a 3.3 V supply at 25 °C of ambient temperature at the RF port. All transmitters' measurements are based on a 100% duty cycle.

<sup>2</sup> The current consumption figures for in RX mode are for cases when the peripherals are disabled and the CPU idle.

Table 9: Current Consumption Depending on Work Modes

| Work Mode                   | Description                                          |                      | Current Consumption (Typ) |
|-----------------------------|------------------------------------------------------|----------------------|---------------------------|
| Modem-sleep <sup>1, 2</sup> | The CPU is powered on <sup>3</sup>                   | 240 MHz              | 30 ~ 68 mA                |
|                             |                                                      | 160 MHz              | 27 ~ 44 mA                |
|                             |                                                      | Normal speed: 80 MHz | 20 ~ 31 mA                |
| Light-sleep                 | —                                                    |                      | 0.8 mA                    |
| Deep-sleep                  | The ULP coprocessor is powered on <sup>4</sup>       |                      | 150 $\mu$ A               |
|                             | ULP sensor-monitored pattern <sup>5</sup>            |                      | 100 $\mu$ A @1% duty      |
|                             | RTC timer + RTC memory                               |                      | 10 $\mu$ A                |
|                             | RTC timer only                                       |                      | 5 $\mu$ A                 |
| Power off                   | CHIP_PU is set to low level, the chip is powered off |                      | 1 $\mu$ A                 |

<sup>1</sup> The current consumption figures in Modem-sleep mode are for cases where the CPU is powered on and the cache idle.

<sup>2</sup> When Wi-Fi is enabled, the chip switches between Active and Modem-sleep modes. Therefore, current consumption changes accordingly.

<sup>3</sup> In Modem-sleep mode, the CPU frequency changes automatically. The frequency depends on the CPU load and the peripherals used.

<sup>4</sup> During Deep-sleep, when the ULP coprocessor is powered on, peripherals such as GPIO and RTC I2C are able to operate.

<sup>5</sup> The “ULP sensor-monitored pattern” refers to the mode where the ULP coprocessor or the sensor works periodically. When ADC works with a duty cycle of 1%, the typical current consumption is 100  $\mu$ A.

## 4.5 Wi-Fi RF Characteristics

### 4.5.1 Wi-Fi RF Standards

Table 10: Wi-Fi RF Standards

| Name                                                     |        | Description                                                                                          |
|----------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------|
| Center frequency range of operating channel <sup>1</sup> |        | 2412 ~ 2484 MHz                                                                                      |
| Wi-Fi wireless standard                                  |        | IEEE 802.11b/g/n                                                                                     |
| Data rate                                                | 20 MHz | 11b: 1, 2, 5.5 and 11 Mbps<br>11g: 6, 9, 12, 18, 24, 36, 48, 54 Mbps<br>11n: MCS0-7, 72.2 Mbps (Max) |
|                                                          | 40 MHz | 11n: MCS0-7, 150 Mbps (Max)                                                                          |
| Antenna type                                             |        | On-board PCB antenna, external antenna <sup>2</sup>                                                  |

<sup>1</sup> Device should operate in the center frequency range allocated by regional regulatory authorities. Target center frequency range is configurable by software.

<sup>2</sup> For the modules that use external antennas, the output impedance is 50  $\Omega$ . For other modules without external antennas, the output impedance is irrelevant.

### 4.5.2 Transmitter Characteristics

Target TX power is configurable based on device or certification requirements. The default characteristics are provided in Table 11.

**Table 11: TX Power Characteristics**

| Rate            | Typ (dBm) |
|-----------------|-----------|
| 11b, 1 Mbps     | 19.5      |
| 11b, 11 Mbps    | 19.5      |
| 11g, 6 Mbps     | 18        |
| 11g, 54 Mbps    | 14        |
| 11n, HT20, MCS0 | 18        |
| 11n, HT20, MCS7 | 13        |
| 11n, HT40, MCS0 | 18        |
| 11n, HT40, MCS7 | 13        |

### 4.5.3 Receiver Characteristics

**Table 12: RX Sensitivity Characteristics**

| Rate            | Typ (dBm) |
|-----------------|-----------|
| 1 Mbps          | -97       |
| 2 Mbps          | -94       |
| 5.5 Mbps        | -92       |
| 11 Mbps         | -88       |
| 6 Mbps          | -93       |
| 9 Mbps          | -91       |
| 12 Mbps         | -89       |
| 18 Mbps         | -87       |
| 24 Mbps         | -84       |
| 36 Mbps         | -80       |
| 48 Mbps         | -77       |
| 54 Mbps         | -75       |
| 11n, HT20, MCS0 | -92       |
| 11n, HT20, MCS1 | -88       |
| 11n, HT20, MCS2 | -86       |
| 11n, HT20, MCS3 | -83       |
| 11n, HT20, MCS4 | -80       |
| 11n, HT20, MCS5 | -76       |
| 11n, HT20, MCS6 | -74       |
| 11n, HT20, MCS7 | -72       |
| 11n, HT40, MCS0 | -89       |
| 11n, HT40, MCS1 | -85       |
| 11n, HT40, MCS2 | -83       |
| 11n, HT40, MCS3 | -80       |
| 11n, HT40, MCS4 | -76       |
| 11n, HT40, MCS5 | -72       |
| 11n, HT40, MCS6 | -71       |

Cont'd on next page

Table 12 – cont'd from previous page

| Rate            | Typ (dBm) |
|-----------------|-----------|
| 11n, HT40, MCS7 | –69       |

Table 13: RX Maximum Input Level

| Rate            | Typ (dBm) |
|-----------------|-----------|
| 11b, 1 Mbps     | 5         |
| 11b, 11 Mbps    | 5         |
| 11g, 6 Mbps     | 0         |
| 11g, 54 Mbps    | –8        |
| 11n, HT20, MCS0 | 0         |
| 11n, HT20, MCS7 | –8        |
| 11n, HT40, MCS0 | 0         |
| 11n, HT40, MCS7 | –8        |

Table 14: Adjacent Channel Rejection

| Rate            | Typ (dB) |
|-----------------|----------|
| 11b, 11 Mbps    | 35       |
| 11g, 6 Mbps     | 27       |
| 11g, 54 Mbps    | 13       |
| 11n, HT20, MCS0 | 27       |
| 11n, HT20, MCS7 | 12       |
| 11n, HT40, MCS0 | 16       |
| 11n, HT40, MCS7 | 7        |

## 4.6 Bluetooth Radio

### 4.6.1 Receiver – Basic Data Rate

Table 15: Receiver Characteristics – Basic Data Rate

| Parameter                         | Conditions     | Min | Typ | Max | Unit |
|-----------------------------------|----------------|-----|-----|-----|------|
| Sensitivity @0.1% BER             | —              | –90 | –89 | –88 | dBm  |
| Maximum received signal @0.1% BER | —              | 0   | —   | —   | dBm  |
| Co-channel C/I                    | —              | —   | +7  | —   | dB   |
| Adjacent channel selectivity C/I  | F = F0 + 1 MHz | —   | —   | –6  | dB   |
|                                   | F = F0 – 1 MHz | —   | —   | –6  | dB   |
|                                   | F = F0 + 2 MHz | —   | —   | –25 | dB   |
|                                   | F = F0 – 2 MHz | —   | —   | –33 | dB   |
|                                   | F = F0 + 3 MHz | —   | —   | –25 | dB   |
|                                   | F = F0 – 3 MHz | —   | —   | –45 | dB   |

Cont'd on next page

Table 15 – cont'd from previous page

| Parameter                        | Conditions          | Min | Typ | Max | Unit |
|----------------------------------|---------------------|-----|-----|-----|------|
| Out-of-band blocking performance | 30 MHz ~ 2000 MHz   | -10 | —   | —   | dBm  |
|                                  | 2000 MHz ~ 2400 MHz | -27 | —   | —   | dBm  |
|                                  | 2500 MHz ~ 3000 MHz | -27 | —   | —   | dBm  |
|                                  | 3000 MHz ~ 12.5 GHz | -10 | —   | —   | dBm  |
| Intermodulation                  | —                   | -36 | —   | —   | dBm  |

#### 4.6.2 Transmitter – Basic Data Rate

Table 16: Transmitter Characteristics – Basic Data Rate

| Parameter                         | Conditions       | Min | Typ  | Max | Unit           |
|-----------------------------------|------------------|-----|------|-----|----------------|
| RF transmit power*                | —                | —   | 0    | —   | dBm            |
| Gain control step                 | —                | —   | 3    | —   | dB             |
| RF power control range            | —                | -12 | —    | +9  | dBm            |
| +20 dB bandwidth                  | —                | —   | 0.9  | —   | MHz            |
| Adjacent channel transmit power   | F = F0 ± 2 MHz   | —   | -55  | —   | dBm            |
|                                   | F = F0 ± 3 MHz   | —   | -55  | —   | dBm            |
|                                   | F = F0 ± > 3 MHz | —   | -59  | —   | dBm            |
| $\Delta f_{1avg}$                 | —                | —   | —    | 155 | kHz            |
| $\Delta f_{2max}$                 | —                | 127 | —    | —   | kHz            |
| $\Delta f_{2avg}/\Delta f_{1avg}$ | —                | —   | 0.92 | —   | —              |
| ICFT                              | —                | —   | -7   | —   | kHz            |
| Drift rate                        | —                | —   | 0.7  | —   | kHz/50 $\mu$ s |
| Drift (DH1)                       | —                | —   | 6    | —   | kHz            |
| Drift (DH5)                       | —                | —   | 6    | —   | kHz            |

\* There are a total of eight power levels from 0 to 7, and the transmit power ranges from -12 dBm to 9 dBm. When the power level rises by 1, the transmit power increases by 3 dB. Power level 4 is used by default and the corresponding transmit power is 0 dBm.

#### 4.6.3 Receiver – Enhanced Data Rate

Table 17: Receiver Characteristics – Enhanced Data Rate

| Parameter                          | Conditions     | Min | Typ | Max | Unit |
|------------------------------------|----------------|-----|-----|-----|------|
| $\pi/4$ DQPSK                      |                |     |     |     |      |
| Sensitivity @0.01% BER             | —              | -90 | -89 | -88 | dBm  |
| Maximum received signal @0.01% BER | —              | —   | 0   | —   | dBm  |
| Co-channel C/I                     | —              | —   | 11  | —   | dB   |
| Adjacent channel selectivity C/I   | F = F0 + 1 MHz | —   | -7  | —   | dB   |
|                                    | F = F0 - 1 MHz | —   | -7  | —   | dB   |
|                                    | F = F0 + 2 MHz | —   | -25 | —   | dB   |
|                                    | F = F0 - 2 MHz | —   | -35 | —   | dB   |

Cont'd on next page

Table 17 – cont'd from previous page

| Parameter                          | Conditions                | Min | Typ | Max | Unit |
|------------------------------------|---------------------------|-----|-----|-----|------|
|                                    | $F = F_0 + 3 \text{ MHz}$ | —   | -25 | —   | dB   |
|                                    | $F = F_0 - 3 \text{ MHz}$ | —   | -45 | —   | dB   |
| 8DPSK                              |                           |     |     |     |      |
| Sensitivity @0.01% BER             | —                         | -84 | -83 | -82 | dBm  |
| Maximum received signal @0.01% BER | —                         | —   | -5  | —   | dBm  |
| C/I c-channel                      | —                         | —   | 18  | —   | dB   |
| Adjacent channel selectivity C/I   | $F = F_0 + 1 \text{ MHz}$ | —   | 2   | —   | dB   |
|                                    | $F = F_0 - 1 \text{ MHz}$ | —   | 2   | —   | dB   |
|                                    | $F = F_0 + 2 \text{ MHz}$ | —   | -25 | —   | dB   |
|                                    | $F = F_0 - 2 \text{ MHz}$ | —   | -25 | —   | dB   |
|                                    | $F = F_0 + 3 \text{ MHz}$ | —   | -25 | —   | dB   |
|                                    | $F = F_0 - 3 \text{ MHz}$ | —   | -38 | —   | dB   |

#### 4.6.4 Transmitter – Enhanced Data Rate

Table 18: Transmitter Characteristics – Enhanced Data Rate

| Parameter                                   | Conditions                    | Min | Typ   | Max | Unit |
|---------------------------------------------|-------------------------------|-----|-------|-----|------|
| RF transmit power (see note under Table 16) | —                             | —   | 0     | —   | dBm  |
| Gain control step                           | —                             | —   | 3     | —   | dB   |
| RF power control range                      | —                             | -12 | —     | +9  | dBm  |
| $\pi/4$ DQPSK max $w_0$                     | —                             | —   | -0.72 | —   | kHz  |
| $\pi/4$ DQPSK max $w_i$                     | —                             | —   | -6    | —   | kHz  |
| $\pi/4$ DQPSK max $ w_i + w_0 $             | —                             | —   | -7.42 | —   | kHz  |
| 8DPSK max $w_0$                             | —                             | —   | 0.7   | —   | kHz  |
| 8DPSK max $w_i$                             | —                             | —   | -9.6  | —   | kHz  |
| 8DPSK max $ w_i + w_0 $                     | —                             | —   | -10   | —   | kHz  |
| $\pi/4$ DQPSK modulation accuracy           | RMS DEVM                      | —   | 4.28  | —   | %    |
|                                             | 99% DEVM                      | —   | 100   | —   | %    |
|                                             | Peak DEVM                     | —   | 13.3  | —   | %    |
| 8 DPSK modulation accuracy                  | RMS DEVM                      | —   | 5.8   | —   | %    |
|                                             | 99% DEVM                      | —   | 100   | —   | %    |
|                                             | Peak DEVM                     | —   | 14    | —   | %    |
| In-band spurious emissions                  | $F = F_0 \pm 1 \text{ MHz}$   | —   | -46   | —   | dBm  |
|                                             | $F = F_0 \pm 2 \text{ MHz}$   | —   | -44   | —   | dBm  |
|                                             | $F = F_0 \pm 3 \text{ MHz}$   | —   | -49   | —   | dBm  |
|                                             | $F = F_0 \pm > 3 \text{ MHz}$ | —   | —     | -53 | dBm  |
| EDR differential phase coding               | —                             | —   | 100   | —   | %    |

## 4.7 Bluetooth LE Radio

### 4.7.1 Receiver

Table 19: Receiver Characteristics – Bluetooth LE

| Parameter                          | Conditions                | Min | Typ | Max | Unit |
|------------------------------------|---------------------------|-----|-----|-----|------|
| Sensitivity @30.8% PER             | —                         | −94 | −93 | −92 | dBm  |
| Maximum received signal @30.8% PER | —                         | 0   | —   | —   | dBm  |
| Co-channel C/I                     | —                         | —   | +10 | —   | dB   |
| Adjacent channel selectivity C/I   | $F = F_0 + 1 \text{ MHz}$ | —   | −5  | —   | dB   |
|                                    | $F = F_0 - 1 \text{ MHz}$ | —   | −5  | —   | dB   |
|                                    | $F = F_0 + 2 \text{ MHz}$ | —   | −25 | —   | dB   |
|                                    | $F = F_0 - 2 \text{ MHz}$ | —   | −35 | —   | dB   |
|                                    | $F = F_0 + 3 \text{ MHz}$ | —   | −25 | —   | dB   |
|                                    | $F = F_0 - 3 \text{ MHz}$ | —   | −45 | —   | dB   |
| Out-of-band blocking performance   | 30 MHz ~ 2000 MHz         | −10 | —   | —   | dBm  |
|                                    | 2000 MHz ~ 2400 MHz       | −27 | —   | —   | dBm  |
|                                    | 2500 MHz ~ 3000 MHz       | −27 | —   | —   | dBm  |
|                                    | 3000 MHz ~ 12.5 GHz       | −10 | —   | —   | dBm  |
| Intermodulation                    | —                         | −36 | —   | —   | dBm  |

#### 4.7.2 Transmitter

Table 20: Transmitter Characteristics – Bluetooth LE

| Parameter                                       | Conditions                    | Min | Typ   | Max | Unit                 |
|-------------------------------------------------|-------------------------------|-----|-------|-----|----------------------|
| RF transmit power (see note under Table 16)     | —                             | —   | 0     | —   | dBm                  |
| Gain control step                               | —                             | —   | 3     | —   | dB                   |
| RF power control range                          | —                             | −12 | —     | +9  | dBm                  |
| Adjacent channel transmit power                 | $F = F_0 \pm 2 \text{ MHz}$   | —   | −55   | —   | dBm                  |
|                                                 | $F = F_0 \pm 3 \text{ MHz}$   | —   | −57   | —   | dBm                  |
|                                                 | $F = F_0 \pm > 3 \text{ MHz}$ | —   | −59   | —   | dBm                  |
| $\Delta f_{1\text{avg}}$                        | —                             | —   | —     | 265 | kHz                  |
| $\Delta f_{2\text{max}}$                        | —                             | 210 | —     | —   | kHz                  |
| $\Delta f_{2\text{avg}}/\Delta f_{1\text{avg}}$ | —                             | —   | +0.92 | —   | —                    |
| ICFT                                            | —                             | —   | −10   | —   | kHz                  |
| Drift rate                                      | —                             | —   | 0.7   | —   | kHz/50 $\mu\text{s}$ |
| Drift                                           | —                             | —   | 2     | —   | kHz                  |

## 5 Module Schematics

This is the reference design of the module.

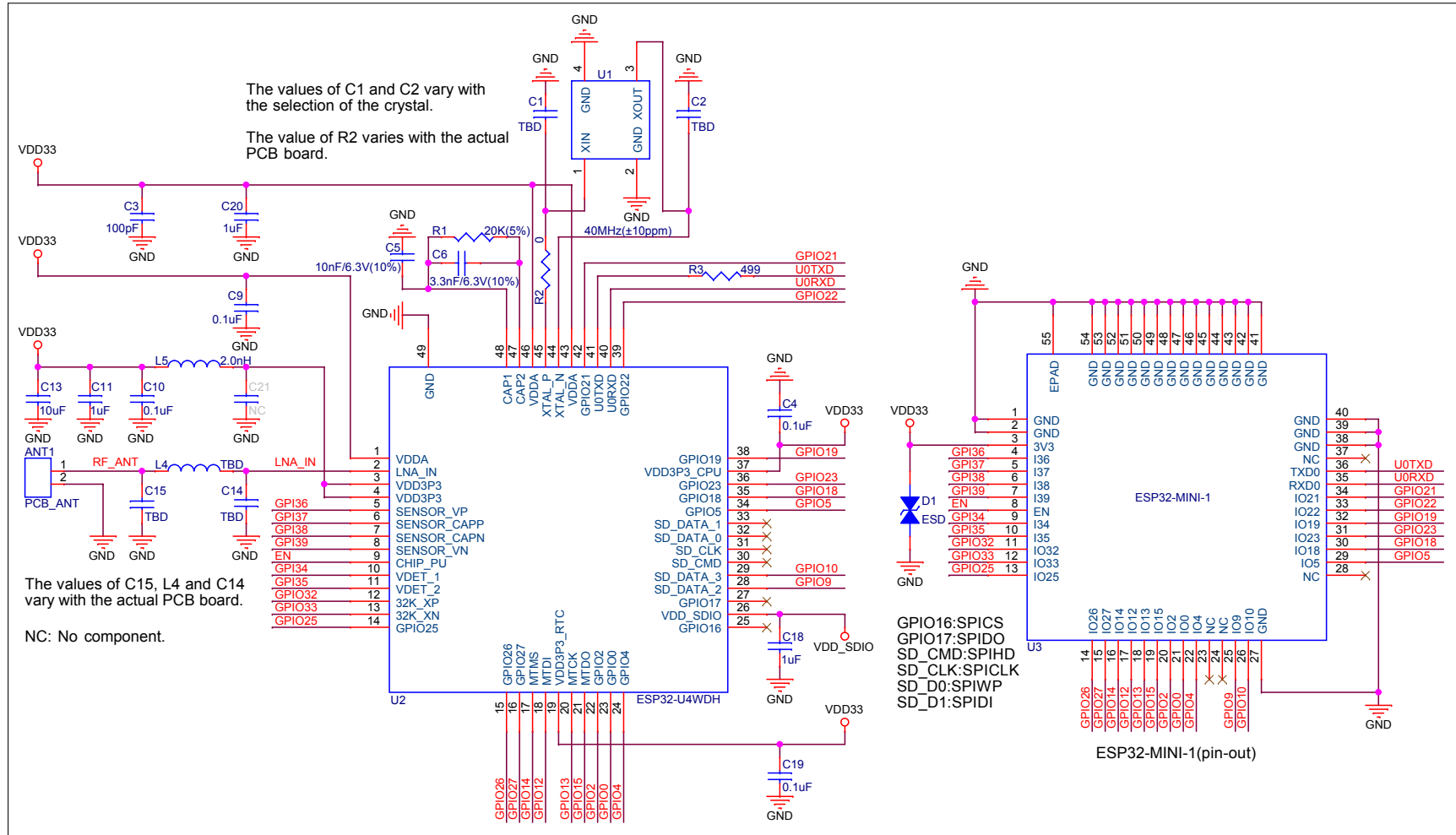


Figure 6: ESP32-MINI-1 Schematics

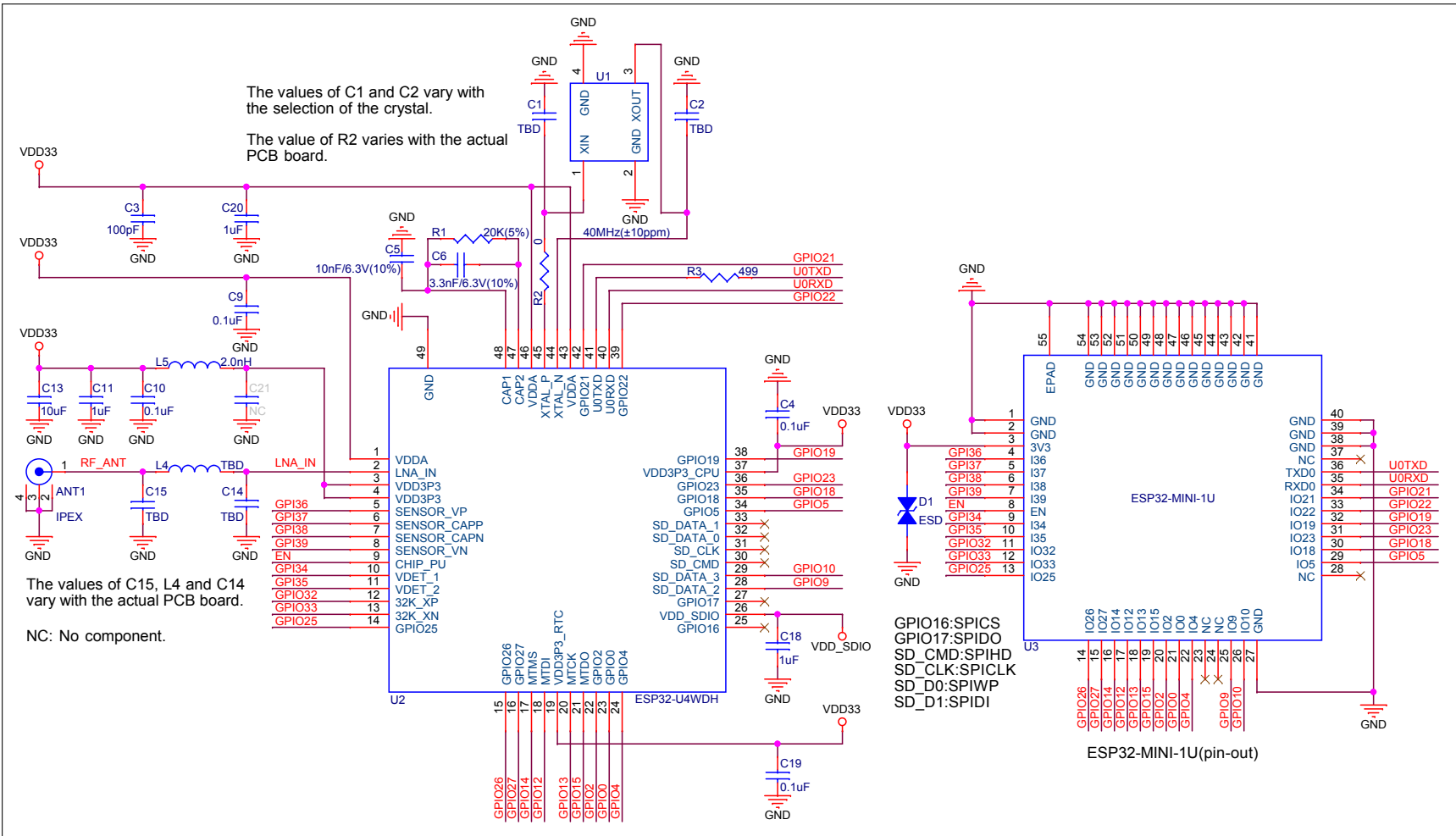


Figure 7: ESP32-MINI-1U Schematics

## 6 Peripheral Schematics

This is the typical application circuit of the module connected with peripheral components (for example, power supply, antenna, reset button, JTAG interface, and UART interface).

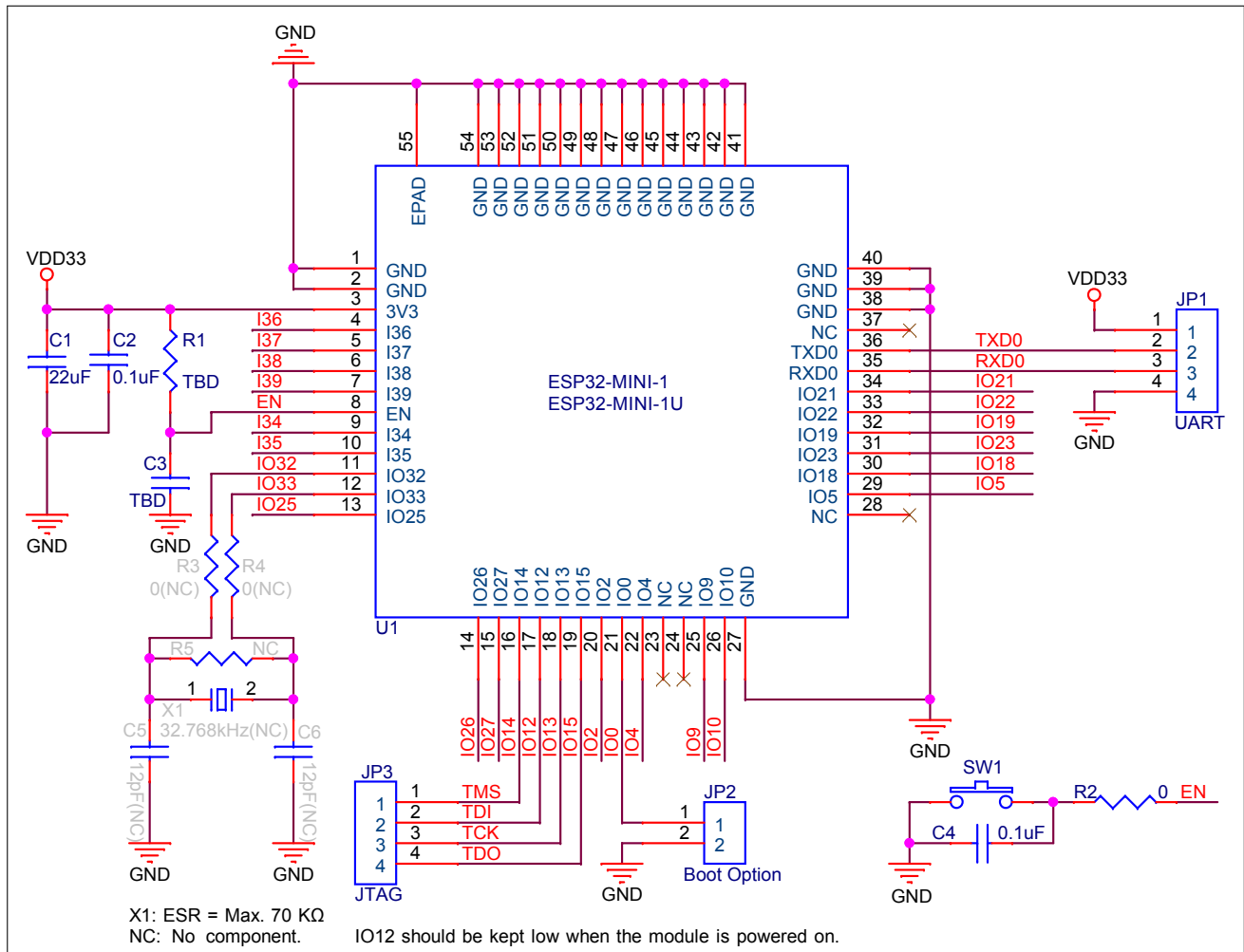


Figure 8: Peripheral Schematics

### Note:

- Soldering Pad 55 to the Ground of the base board is not necessary for a satisfactory thermal performance. If you choose to solder it, please apply the correct amount of soldering paste. Too much soldering paste may increase the gap between the module and the baseboard. As a result, the adhesion between other pins and the baseboard may be poor.
- To ensure that the power supply to the ESP32 chip is stable during power-up, it is advised to add an RC delay circuit at the EN pin. The recommended setting for the RC delay circuit is usually  $R = 10\text{ k}\Omega$  and  $C = 1\text{ }\mu\text{F}$ . However, specific parameters should be adjusted based on the power-up timing of the module and the power-up and reset sequence timing of the chip. For ESP32's power-up and reset sequence timing diagram, please refer to Section Power Scheme in [ESP32 Series Datasheet](#).

## 7 Physical Dimensions and PCB Land Pattern

### 7.1 Physical Dimensions

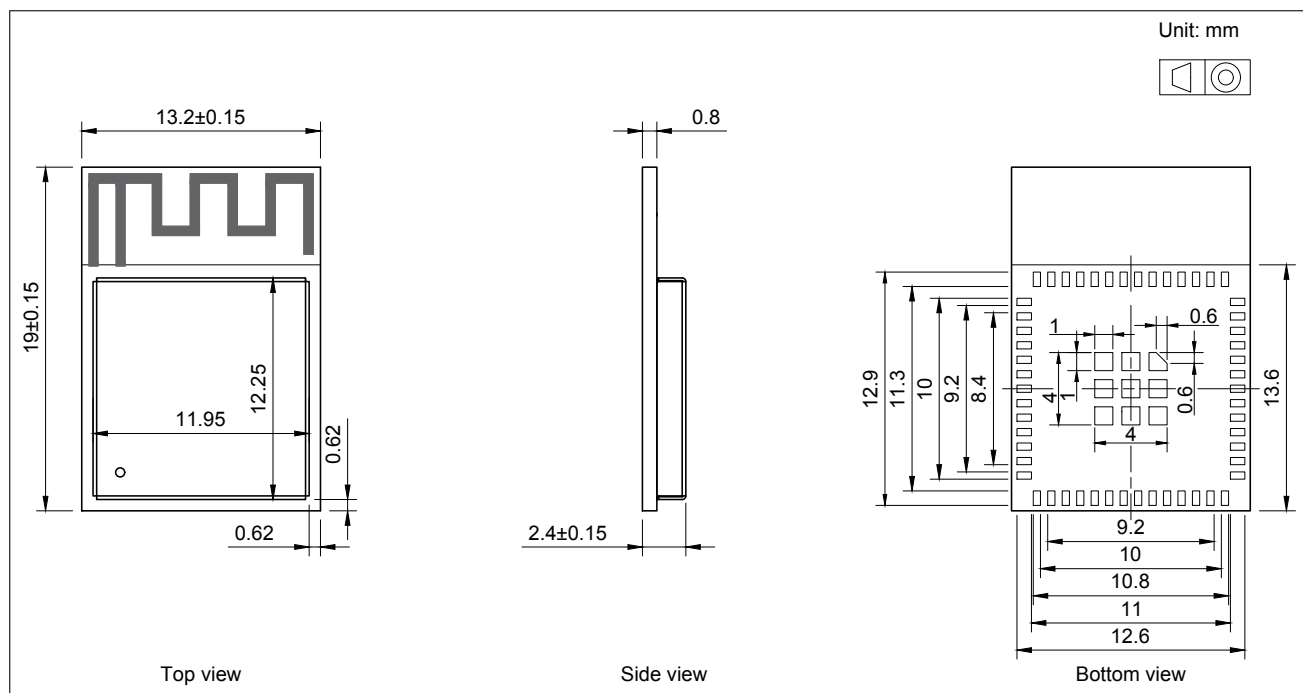


Figure 9: ESP32-MINI-1 Physical Dimensions

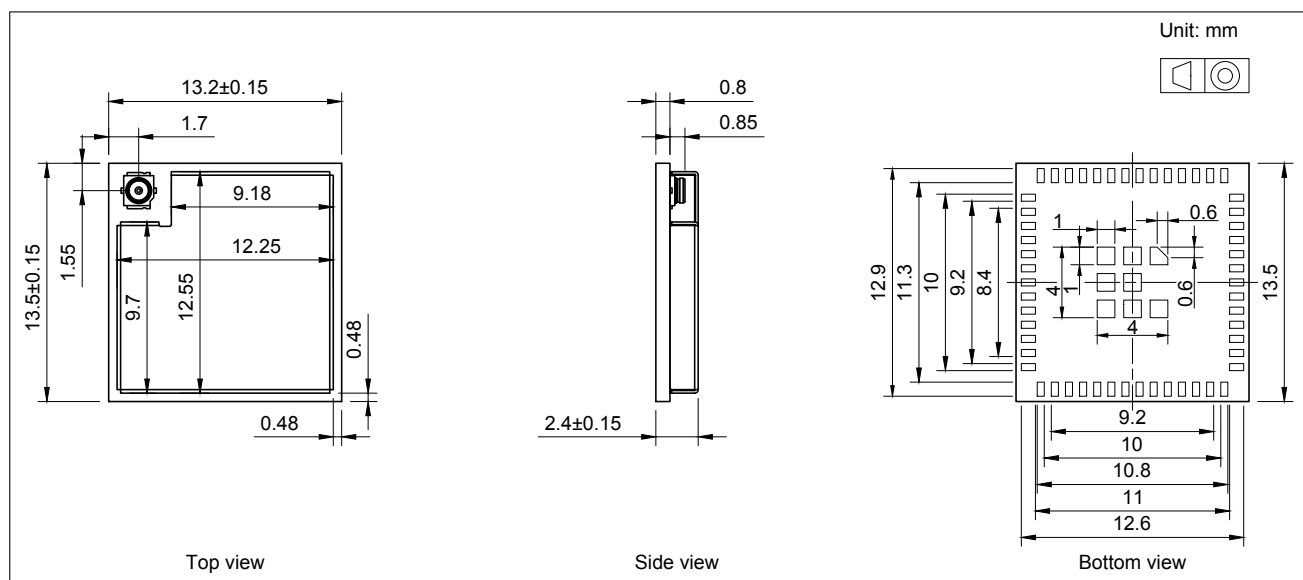


Figure 10: ESP32-MINI-1U Physical Dimensions

**Note:**

For information about tape, reel, and product marking, please refer to [Espressif Module Package Information](#).

## 7.2 Recommended PCB Land Pattern

This section provides the following resources for your reference:

- Figures for recommended PCB land patterns with all the dimensions needed for PCB design. See Figure 11 *ESP32-MINI-1 Recommended PCB Land Pattern* and Figure 12 *ESP32-MINI-1U PCB Recommended PCB Land Pattern*.
- Source files of recommended PCB land patterns to measure dimensions not covered in Figure 11 and Figure 12. You can view the source files for [ESP32-MINI-1](#) and [ESP32-MINI-1U](#) with [Autodesk Viewer](#).
- 3D models of [ESP32-MINI-1](#) and [ESP32-MINI-1U](#). Please make sure that you download the 3D model file in .STEP format (beware that some browsers might add .txt).

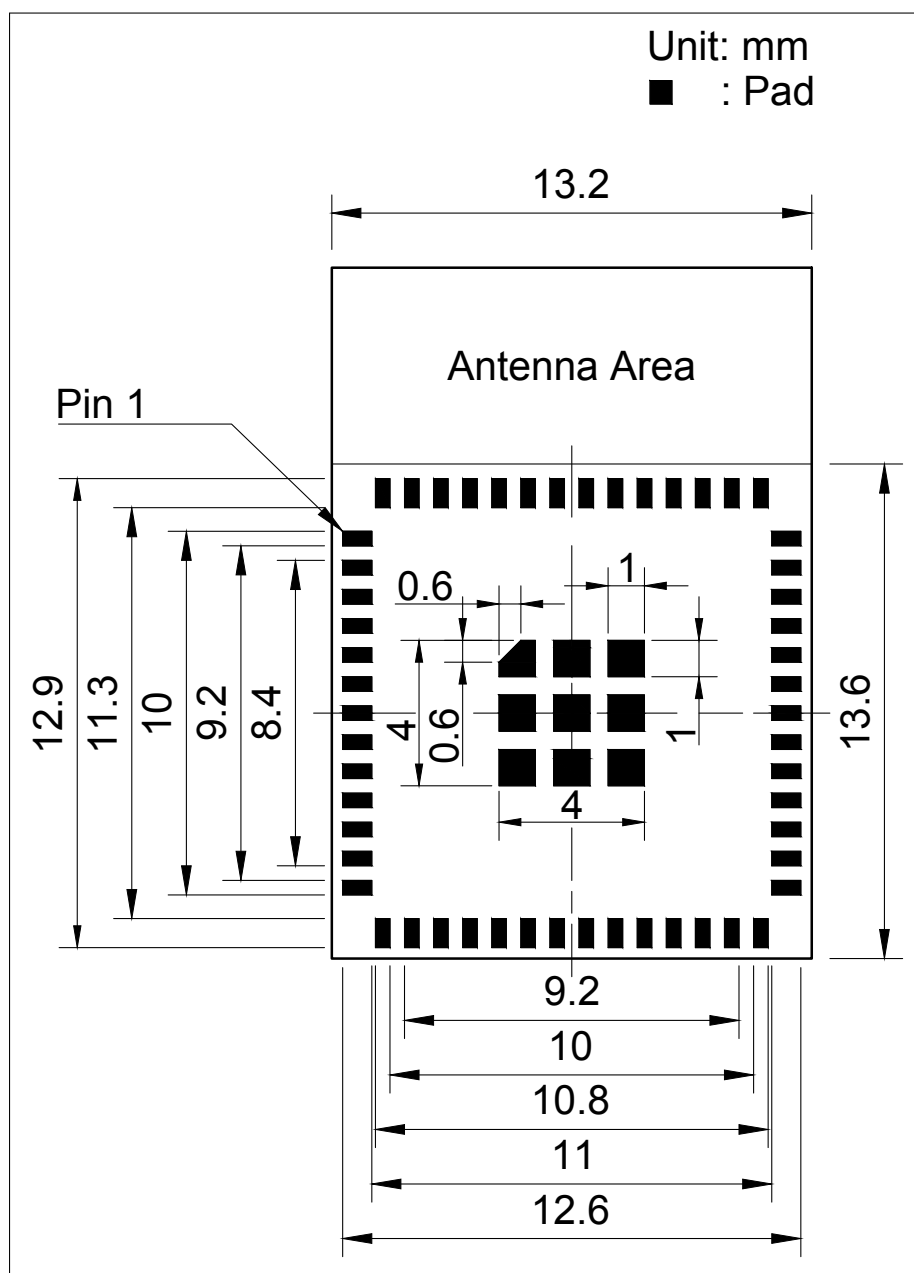


Figure 11: ESP32-MINI-1 Recommended PCB Land Pattern

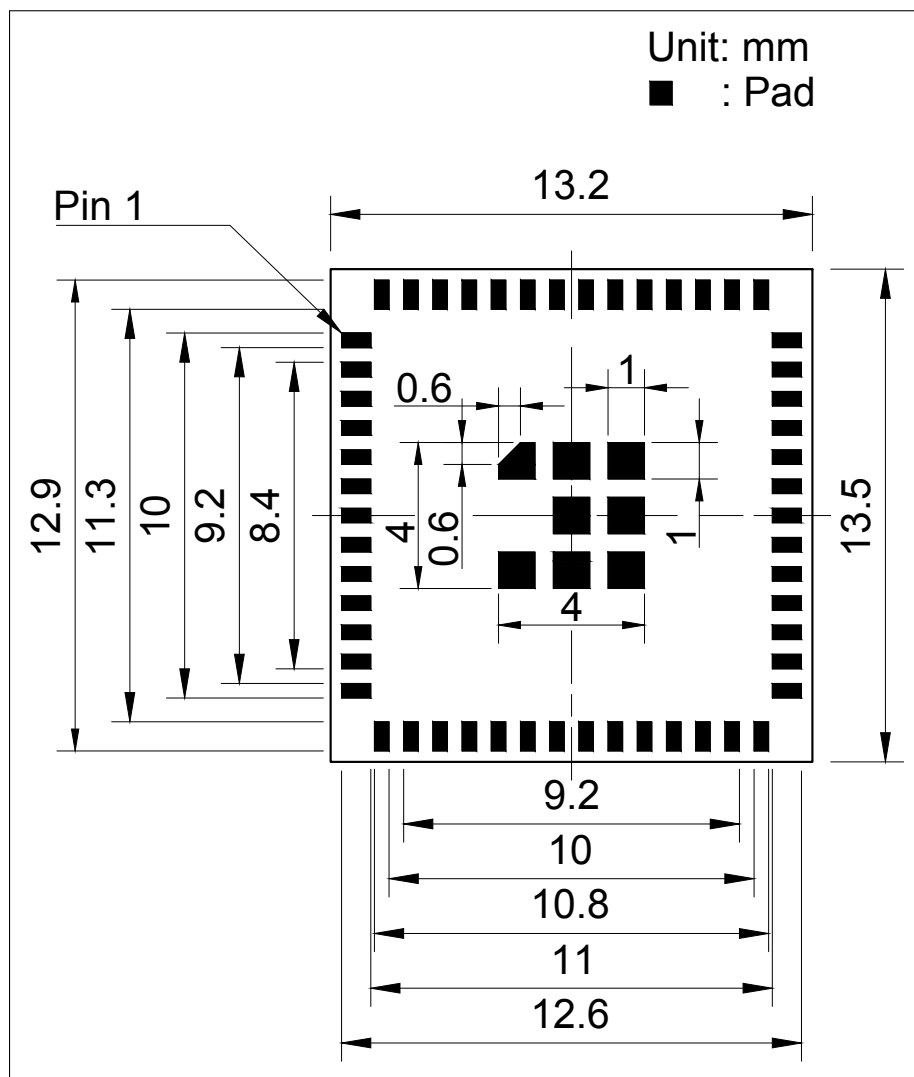


Figure 12: ESP32-MINI-1U PCB Recommended PCB Land Pattern

### 7.3 Dimensions of External Antenna Connector

ESP32-MINI-1U uses the third generation external antenna connector as shown in Figure 13. This connector is compatible with the following connectors:

- W.FL Series connector from Hirose
- MHF III connector from I-PEX
- AMMC connector from Amphenol

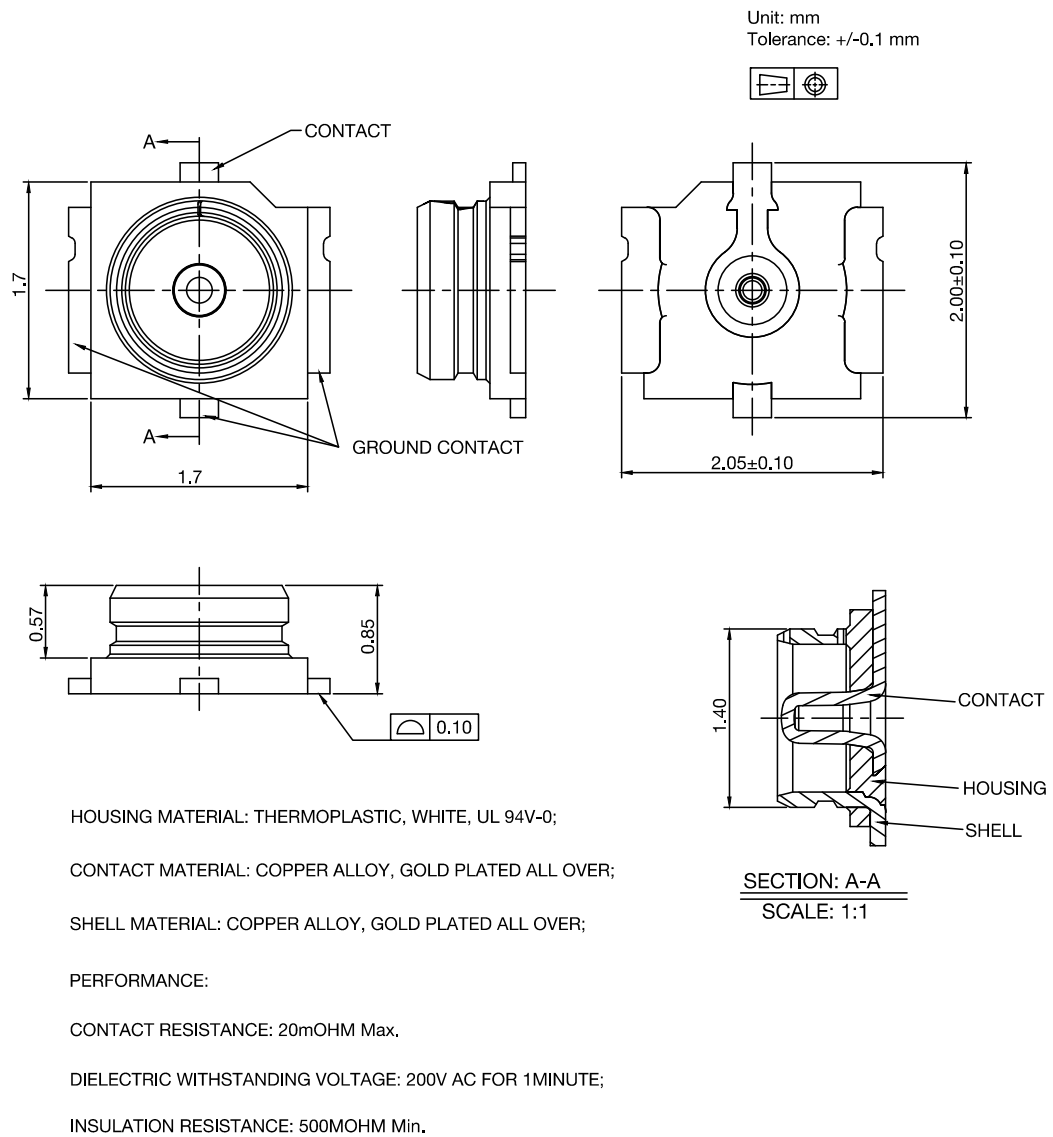


Figure 13: Dimensions of External Antenna Connector

## 8 Product Handling

### 8.1 Storage Conditions

The products sealed in moisture barrier bags (MBB) should be stored in a non-condensing atmospheric environment of  $< 40\text{ }^{\circ}\text{C}$  and 90%RH. The module is rated at the moisture sensitivity level (MSL) of 3.

After unpacking, the module must be soldered within 168 hours with the factory conditions  $25 \pm 5\text{ }^{\circ}\text{C}$  and 60 %RH. If the above conditions are not met, the module needs to be baked.

### 8.2 Electrostatic Discharge (ESD)

- Human body model (HBM):  $\pm 2000\text{ V}$
- Charged-device model (CDM):  $\pm 500\text{ V}$

### 8.3 Reflow Profile

Solder the module in a single reflow.

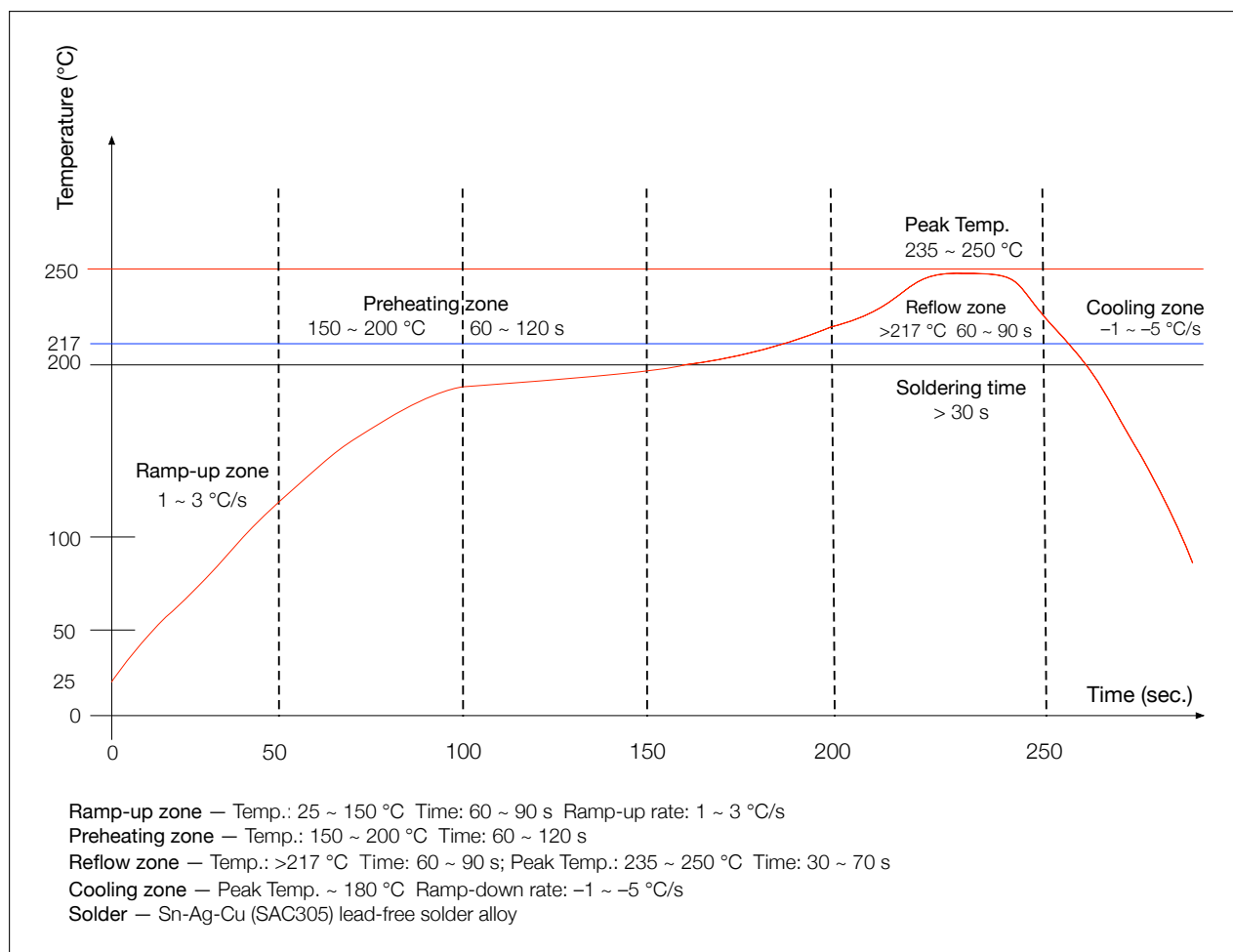


Figure 14: Reflow Profile

## 8.4 Ultrasonic Vibration

Avoid exposing Espressif modules to vibration from ultrasonic equipment, such as ultrasonic welders or ultrasonic cleaners. This vibration may induce resonance in the in-module crystal and lead to its malfunction or even failure. As a consequence, **the module may stop working or its performance may deteriorate.**

## 9 Related Documentation and Resources

### Related Documentation

- [ESP32 Series Datasheet](#) – Specifications of the ESP32 hardware.
- [ESP32 Technical Reference Manual](#) – Detailed information on how to use the ESP32 memory and peripherals.
- [ESP32 Hardware Design Guidelines](#) – Guidelines on how to integrate the ESP32 into your hardware product.
- [ESP32 ECO and Workarounds for Bugs](#) – Correction of ESP32 design errors.
- *Certificates*  
<https://espressif.com/en/support/documents/certificates>
- *ESP32 Product/Process Change Notifications (PCN)*  
<https://espressif.com/en/support/documents/pcns>
- *ESP32 Advisories* – Information on security, bugs, compatibility, component reliability.  
<https://espressif.com/en/support/documents/advisories>
- *Documentation Updates and Update Notification Subscription*  
<https://espressif.com/en/support/download/documents>

### Developer Zone

- [ESP-IDF Programming Guide for ESP32](#) – Extensive documentation for the ESP-IDF development framework.
- *ESP-IDF* and other development frameworks on GitHub.  
<https://github.com/espressif>
- *ESP32 BBS Forum* – Engineer-to-Engineer (E2E) Community for Espressif products where you can post questions, share knowledge, explore ideas, and help solve problems with fellow engineers.  
<https://esp32.com/>
- *The ESP Journal* – Best Practices, Articles, and Notes from Espressif folks.  
<https://blog.espressif.com/>
- See the tabs *SDKs and Demos*, *Apps*, *Tools*, *AT Firmware*.  
<https://espressif.com/en/support/download/sdks-demos>

### Products

- *ESP32 Series SoCs* – Browse through all ESP32 SoCs.  
<https://espressif.com/en/products/socs?id=ESP32>
- *ESP32 Series Modules* – Browse through all ESP32-based modules.  
<https://espressif.com/en/products/modules?id=ESP32>
- *ESP32 Series DevKits* – Browse through all ESP32-based devkits.  
<https://espressif.com/en/products/devkits?id=ESP32>
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<https://products.espressif.com/#/product-selector?language=en>

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## Revision History

| Date       | Version | Release Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023-08-28 | v1.3    | <ul style="list-style-type: none"> <li>Table 2 <i>Pin Definitions</i>: Updated the first note about chip pins connecting flash</li> <li>Section 6 <i>Peripheral Schematics</i>: Updated the first note about EPAD soldering</li> <li>Section 7.2 <i>Recommended PCB Land Pattern</i>: Added the 3D file for ESP32-MINI-1U</li> </ul>                                                                                                                                                                  |
| 2023-01-17 | v1.2    | <p>Major updates:</p> <ul style="list-style-type: none"> <li>Removed contents about hall sensor according to <a href="#">PCN20221202</a></li> <li>Added Section 8.4: <i>Ultrasonic Vibration</i></li> </ul> <p>Other updates:</p> <ul style="list-style-type: none"> <li>Added strapping pin timing in Section 3.3: <i>Strapping Pins</i></li> <li>Added source files of PCB land patterns and 3D models of the modules (if available) in Section 7.2: <i>Recommended PCB Land Pattern</i></li> </ul> |
| 2021-11-09 | v1.1    | <p>Upgraded the module embedded chip from single-core to dual-core<sup>1</sup></p> <p>Updated the description to TWAI</p> <p>Updated Table 6: <i>Recommended Operating Conditions</i></p>                                                                                                                                                                                                                                                                                                             |
| 2021-07-14 | v1.0    | <p>Added ESP32-MINI-1U module</p> <p>Updated the document formatting</p>                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 2020-12-04 | v0.5    | Pre-release                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

<sup>1</sup>The module embedded chip (ESP32-U4WDH) is upgraded from single-core to dual-core, see [PCN-2021-021](#). Estimated effective date: January 1, 2022. The single-core version coexists with the new dual-core version around January 1, 2022. The physical product is subject to batch tracking.



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